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Technology Review

Podcasting for instructional effectiveness

Jim Pham

Alliant International University

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Today's students operate in a digital and global environment dominated by digital devices—cell phones, personal digital assistants (PDAs), and mobile devices—and so does today's business (Coleman, 2005). Virtually every electronic device can connect directly to the Internet. Many college students already have access to the Internet through their schools or commercial Internet service providers. Many of today's popular Internet-capable devices include cell phones, laptop computers, PDAs, the iPod, the iPhone, and various digital media players. Digital media players, such as iPods/iPhone have been very popular among college students because they can serve as an entertainment center (music, video, games, etc.) and a learning device. Many teaching methods can also be enhanced using digital media devices and podcasting, helping educators become more effective in their teaching. Virtually all course content, lectures, and assignments can easily be converted to MP3 digital format or other audio/video podcast formats, so students can review the course materials outside of the classroom. This is what educational podcasts are all about. According to Fordham University RETC Center for Professional Development, life is about change, and learning through change can be transformative or constructing meaning (<http://www.podcastforteachers.org/>, http://en.wikipedia.org/wiki/Transformative_learning/). A Podcast is about embracing change in teaching and learning, and it reflects a type of transformative learning theory most frequently found in constructivist approaches to education.

Podcasts are derived from Apple's portable music player, the iPod. The iPod and Podcast technologies can deliver educational content to students. Students can play podcasts directly on iTunes (Apple media player) or with a computer media player or any MP3 or digital media player (<http://en.wikipedia.org/wiki/Podcast>).

Podcasting is an audio/video file feed in the Really Simple Syndication (RSS) format, a family of web feed formats used to publish frequently updated digital content, such as blogs. Podcasting is also a term used to describe a collection of technologies for automatically distributing audio and video programs over the Internet. Podcasting enables educators to create self-published audio/video content from course material, lectures, and assignments. Most digital media players or computers with audio-playing software can play podcasts. The term podcast, however, still refers largely to audio/video content distribution (<http://en.wikipedia.org/wiki/Podcast>).

How does podcasting change the way teachers teach and students learn? Podcasts allow a teacher to easily create self-published course content and lectures. A Teacher can also listen to his/her own lectures to improve their presentation skills. Students can download and replay any published podcasts by their teacher that have been covered in class. Students can listen to or review any previous lectures outside of the classroom. A Podcast is highly flexible audio/video content that students can listen to at home, on the way to work, or on the go. Listening to podcasts is also a great way to review lectures before an exam.

Podcasts are related to notetaking or reviewing lecture notes. Traditionally, note-taking is a common activity in school settings, and a large percentage of students have been reported to take notes (Palmatier & Bennett, 1974). Cohn, E., Cohn, S., and Bradley (1995) argued that when listening and taking notes, students must attend to the lecturer, abstract important information, and then hold and manipulate a representation of that information in working memory while recording notes. In addition, students are less likely to pay attention to the lecture while taking notes. Kiewra (1991) offers suggestions to facilitate selecting information to record during a lecture. These suggestions include using cueing strategies such

as recording information on the blackboard and repeating the lecture, as well as using a tape or video.

Frey and Birnbaum (2002) have stated that computer-assisted instruction has a positive effect on lectures, especially in taking notes and studying for exams. In addition, research suggests that a technology-based approach to instruction is perceived favorably by students and yields significant improvements in student learning, as evidenced by both student self-reports and objective outcome testing (Smith & Woody, 2000).

The World Wide Web has also created an information-rich teaching and learning environment that facilitates social interaction and exchange among students and instructors. It allows teachers and students to access a wealth of knowledge-based information and assessment tools that replicate and expand the traditional classroom. It has proven effective for storing, disseminating, and retrieving course-relevant information, and it is available to students anytime, anywhere (Aggarwal & Bento, 2000).

Given this varied set of results, an increasing number of teachers are using technology-assisted instruction in their classes. In this research study, educators will have a better understanding of student's use of technology offered by their teacher relative to the more traditional classroom methods of learning. Educational podcasts are a new technology that allows online audio/video communication between students and teachers, delivered via RSS Podcasting and available for students to listen to on their PDAs or personal computers. Instead of reviewing your class notes, why not replay the class lecture directly on students' portable media players via podcasting? This also comes with a cost for both the students and the teacher, as they must have an iPod or MP3 player. In addition, teachers must regularly publish podcasts of the class information.

Many research studies on technology-based learning have also shown that students are likely to focus on the best resources to learn. One such study is by Aytekin (2005), which states that students learn best when they actively construct their own understanding, in addition to reflective and more responsive forms of education. Educational podcasts are a new technology trend that is widely popular, particularly among students.

One study is on instructional technologies that support learning preferences of today's students. Students prefer to use technology to communicate with each other and with their instructor. Communication such as E-mail, discussion boards, instant messaging, blogs, and conference systems have created more efficient and effective learning and teaching environments. With advances in communication, educators have made technology-based learning possible through information technology (Ficek, 2006), including computers and PDAs.

Another study by Frey and Birnbaum (2002) found that technology-based education offers a hands-on approach to improving educational outcomes. Since then, technology has undergone significant advances in delivering educational podcasts via the Internet, digital media players, and iTunes or any other media player software. To address this new way of teaching and learning via educational podcasts—audio/video podcasts morphed into traditional classroom learning—where lectures, notes, and assignments are podcasted over the Internet to support learning outside the classroom.

Currently, there is no study on the effect of educational podcasts on student performance (or learning) or on how podcasts can be used to enhance student performance. Based on all the previous studies on technology-based learning, it is suggested that there is a

correlation between podcasts and student performance. Further study is required to determine how podcasts can be used to enhance student performance.

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